DAD FieldWorks

A Reusable Evidence Contract Architecture for Validation-Aware RF and Quantum-Hardware-Oriented Electromagnetic Design Workflows

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Claim status: No external validation claim, no production readiness claim.

Abstract

DAD FieldWorks is a private research and development project for validation-aware RF and quantum-hardware-oriented electromagnetic design workflows. This public specification draft describes a reusable evidence contract architecture for computational electromagnetics software. The central idea is simple: numerical output should carry evidence state, decision state, failure state, risk state, comparison state, required future work, and explicit claim boundaries.

Version 0.7 updates the public whitepaper to reflect the current public-safe project state. It summarizes records-only evidence infrastructure, bounded native Yee curl incidence and curl-curl prototype status, the PEC cavity eigenmode and residual/reference diagnostic path, the FDTD microwave resonator ringdown and FFT diagnostic, the public showcase visualizations, and the future C++ evidence-contract direction. The document remains a public architecture draft. It is not validation evidence, not a product release, and not an external solver equivalence claim.

Contents

1 Introduction

Computational electromagnetics research often produces useful numerical output before that output is ready to support a public claim. A frequency estimate, residual, field slice, time trace, spectrum, or comparison row can be valuable for engineering research while remaining insufficient for validation, production use, or tool equivalence.

DAD FieldWorks treats that separation as an architecture problem. A result should not carry only numbers. It should carry the evidence context needed to understand what was computed, what was checked, what remains blocked, and what cannot yet be claimed.

Public boundary. This document describes public-safe architecture, status, and visual diagnostics. It does not publish private solver source code, internal test suites, notebooks, private artifact files, private paths, or external organization information.

2 Motivation

Engineering software benefits from a disciplined distinction between computation and evidence. A solver run can be finite and still be inconclusive. A diagnostic spectrum can show a peak and still fail to identify a physical mode. A prototype row record can be internally consistent and still not be a production operator.

The DAD FieldWorks evidence contract architecture is designed to keep those distinctions visible. It asks every result to carry explicit records for source assumptions, geometry assumptions, numerical method state, comparison scope, pass/fail gates, risk factors, missing evidence, and allowed claims. This reduces the chance that a useful internal diagnostic is accidentally presented as a stronger public result.

3 Validation-Aware RF and Quantum-Hardware-Oriented Direction

DAD FieldWorks is being developed toward validation-aware RF and quantum-hardware-oriented electromagnetic design workflows. The project currently addresses classical electromagnetic modeling, RF PCB workflows, microwave structures, resonator and cavity analysis, solver research, and evidence records.

Quantum-hardware orientation currently means microwave resonator, cavity, field-mode, package-mode, feedline, and coupling diagnostics. These are classical electromagnetic structures around which quantum-hardware engineering workflows often need careful RF and microwave analysis.

This orientation does not mean qubit simulation. It does not mean Josephson junction modeling. It does not mean quantum processor validation. It does not imply Hamiltonian extraction, coherence-time prediction, or a complete quantum hardware solver.

4 Evidence Contract Architecture

An evidence contract is a structured result envelope around a computational or diagnostic result. It combines the numerical payload with evidence rows, gate rows, decision rows, failure rows, risk rows, comparison rows, claim-boundary rows, and required-future-work rows.

The architecture separates:

- computation from claim status,
- internal consistency from validation,
- diagnostic usefulness from production readiness,
- future direction from current implementation,
- numerical success from physical acceptance.

Two Boolean claim fields remain central:

- `ExternallyValidatedQ` remains false unless a real comparison evidence path supports changing it.
- `ProductionAllowedQ` remains false unless a separate production release gate supports changing it.

5 Evidence Rows and Gate Rows

Evidence rows describe facts about a result. Typical rows include geometry identity, material assumptions, boundary conditions, source and probe settings, grid conventions, field component availability, finite-value sanity checks, residual values, comparison inputs, and missing-evidence records.

Gate rows describe conditions. They may pass, fail, or remain blocked. A gate row can say that a diagnostic is internally consistent, that a comparison is not yet ready, that a result is finite, or that a stronger claim must remain blocked.

The evidence/gate separation prevents a result from becoming stronger than its record. Evidence rows record what is known. Gate rows say what that evidence permits.

6 Decision, Failure, Risk, and Claim Boundary Rows

Decision rows classify the current result state. Failure rows identify why a result failed or remained blocked. Risk rows record unresolved interpretation issues. Claim-boundary rows state what cannot be claimed from the current evidence.

This pattern is especially important for electromagnetic solvers because many intermediate objects can look convincing. A field slice can look physically plausible while mode identity remains unresolved. A row structure can be consistent while operator closure remains incomplete. A spectrum can have a dominant component while source, probe, and window sensitivity remain material.

7 Reference Comparison Model

Reference comparison is treated as a separate evidence path. The architecture requires explicit rows for reference source, geometry equivalence, material equivalence, boundary equivalence, grid mapping, field component interpretation, comparison metric, uncertainty policy, and acceptance policy.

The current PEC cavity path uses a rectangular cavity reference as a public-safe case study. It separates analytical reference preparation, scalar eigenvalue prototypes, same-grid scalar diagnostics, residual records, and time-domain ringdown diagnostics. The separation is intentional: agreement in one diagnostic layer is not automatically validation of another layer.

8 Current Public Technical Status

The current private project state remains research and development. The public-safe summary is:

- Records-only evidence infrastructure is active and central.
- The PEC cavity evidence chain has expanded from scalar reference and diagnostic rows toward bounded residual and analytical comparison records.
- Bounded native Yee curl incidence and bounded curl-curl prototype layers are records-only and claim-bounded.
- The current internal queue is focused on records-only planning and auditing for a Resonator Lab Alpha report prototype.
- The public site includes solver-generated diagnostic visualizations and public-safe schematic visualizations.
- C++ evidence-contract infrastructure is a future production-core direction and includes contract-oriented result packaging concepts.
- No GUI is currently claimed.
- No production solver is currently claimed.
- No external validation claim is made.

Recent internal records include bounded native Yee curl-curl prototype closure, minimal mass/generalized eigenproblem planning, minimal PEC cavity eigenmode solve prototype work, residual and analytical comparison records, and Resonator Lab Alpha report prototype planning. These records are internal evidence and planning layers; they do not publish private implementation details and do not create validation or production claims.

9 Bounded Native Yee Curl Incidence Microprototype

The bounded native Yee curl incidence microprototype is an operator-near diagnostic layer. It represents how Yee electric unknowns with local orientation can contribute signed entries to local curl incidence rows.

The public visualization shows:

- Yee electric unknowns on oriented local edges,
- a local curl loop orientation,
- plus and minus incidence signs,
- a symbolic bounded row record,
- a claim-boundary card.

The current internal status is conservative. The incidence microprototype has records-only implementation, audit, closure-readiness, closure, and closure-audit layers. Later bounded curl-curl prototype records have also been prepared and closed as records-only layers. These records do not create a production operator.

Microprototype boundary. This is not curl-curl assembly. It is not a production incidence matrix. It is not a matrix eigensolve. It is not production solver behavior. It is not external validation.

10 PEC Cavity Eigenmode Evidence Chain

The PEC cavity case study is used as a controlled evidence-chain example. The public-safe rectangular cavity parameters are:

Quantity	Value
Cavity dimensions	$a = 0.08\,\text{m}$, $b = 0.084\,\text{m}$, $d = 0.084\,\text{m}$
Mode indices	$m = n = p = 1$
Speed of light	$c_0 = 299792458\,\text{m/s}$
Reference frequency	approximately 3.143165987503105 GHz

The scalar Helmholtz refinement diagnostic remains useful as a controlled numerical record:

Level	Grid	Frequency (GHz)	Relative error
0	$\{9, 8, 8\}$	3.128307359058	0.0047272809
1	$\{18, 16, 16\}$	3.139012096545	0.0013215627
2	$\{27, 24, 24\}$	3.141247276062	0.0006104391

The corresponding residual norms are approximately 2.63×10^{-15}, 1.29×10^{-14}, and 1.68×10^{-14}.

The same-grid scalar diagnostic remains claim-bounded:

Policy	Matrix size	Frequency (GHz)	Relative error
1	504×504	2.778104015200	0.1161446687
2	336×336	3.122528486251	0.0065658325
3	504×504	3.126946915965	0.0051601066

The corresponding residual norms are approximately 1.83×10^{-15}, 1.14×10^{-15}, and 2.37×10^{-15}.

The same-grid scalar diagnostic remains inconclusive because mapping policy ambiguity is material. The scalar path does not become a full vector Maxwell eigenmode claim. The later bounded minimal PEC cavity eigenmode solve and residual/reference records remain internal diagnostic evidence unless a future evidence path supports stronger status.

11 FDTD Microwave Resonator Ringdown Diagnostic

The public showcase includes an FDTD microwave resonator ringdown diagnostic. It shows field build-up in a resonator-like geometry, late-time ringdown after a pulse excitation, a time-domain probe signal, and an FFT spectrum derived from that probe signal.

The diagnostic uses a 2D TMz-style field update with signed E_z rendering, a local probe trace, cavity field energy, and a normalized spectrum. The FFT panel is computed from the recorded time-domain probe signal; it is not a hand-drawn spectrum.

This diagnostic is relevant to RF and quantum-hardware-oriented microwave workflows because resonator and feedline behavior are part of classical electromagnetic design around many quantum-hardware contexts. It is not a qubit simulation, not a measured resonance, not validation evidence, and not production readiness.

12 Microwave Cavity Eigenmode Birth Visualization

The public showcase also contains a deterministic public-safe visualization that connects the operator-near evidence chain to a cavity-mode picture. It presents Yee electric unknowns, oriented curl incidence, bounded curl-curl structure visualization, a standing PEC cavity eigenmode slice, residual check, and analytical PEC reference comparison as a claim-bounded communication sequence.

The visualization is intended to help readers understand how local orientation records and bounded diagnostic rows relate to later cavity-mode evidence. It is not a production solver claim, not an external validation claim, and not a qubit simulation.

13 C++ Evidence Contract Direction

C++ remains the intended future production-core direction. The current public-safe direction is evidence-contract infrastructure: result containers, row records, gate state, failure state, and claim boundaries for future solver wrappers.

In public specification terms, this direction includes a future C++ core interface and a future C++ implementation path for mature research kernels once evidence contracts, tests, and claim boundaries are strong enough to justify that migration.

The current public claim is architectural. Non-numerical demo wrappers and evidence-contract structures may demonstrate packaging patterns, but they do not claim production electromagnetic solver behavior. Private C++ source code is not published in this repository.

14 Public Showcase Visualizations

The public site currently features:

- a main solver diagnostic hero animation,
- a PEC cavity eigenmode field-slice visualization,
- a quantum-hardware-oriented microwave resonator field-mode diagnostic,
- an FDTD microwave resonator ringdown diagnostic,
- a Yee curl incidence microprototype visualization,
- an FDTD resonator ringdown and FFT spectrum visualization.

These assets are public-safe diagnostic visualizations. They use local generated assets only. They are not validation evidence, not production readiness, and not commercial solver equivalence.

Retained assets that are not featured on the homepage are not part of the main public visual story. They do not broaden the claim boundary.

15 Limitations and Claim Boundaries

The public claim boundary is deliberately narrow:

- no external validation claim,
- no production readiness claim,
- no commercial solver equivalence claim,
- no full commercial solver claim,
- no arbitrary geometry support claim,
- no CPML support claim,
- no qubit simulation claim,
- no Josephson junction model claim,
- no quantum processor validation claim,
- no complete quantum hardware solver claim,
- no production C++ electromagnetic solver claim,
- no GUI claim,
- no measurement validation claim.

16 Roadmap

Near-term work remains evidence-oriented:

- operator-near Yee incidence closure and evidence records,
- PEC cavity residual, mass, and divergence closure-readiness,
- quantum-hardware-oriented microwave resonator diagnostics,
- whitepaper and public evidence documentation.

Mid-term work remains architecture and policy oriented:

- native Yee curl-curl planning,
- operator validity and residual policy resolution,
- C++ evidence contract integration,
- RF PCB and resonator workflows,
- public-safe diagnostic visualization pipeline.

Long-term work remains conditional on evidence:

- validated reference workflows when evidence supports them,
- external comparison pipelines when ready,
- future GUI only when the core workflow is mature,
- future quantum-hardware-oriented RF design workflows,
- no qubit simulation until actual models and evidence exist.

17 Conclusion

DAD FieldWorks is being developed as a validation-aware research and development project for RF, microwave, and quantum-hardware-oriented electromagnetic workflows. The reusable evidence contract architecture keeps numerical output tied to evidence rows, gate rows, decision rows, failure rows, risk rows, comparison rows, and claim boundaries.

The current public state is useful, but intentionally bounded. The project contains public architecture notes, public whitepaper material, and diagnostic visualizations. It does not publish private solver source code, does not claim external validation, and does not claim production readiness.

18 References

References

[1] Kane S. Yee, "Numerical solution of initial boundary value problems involving Maxwell's equations in isotropic media," *IEEE Transactions on Antennas and Propagation*, 1966.

[2] Constantine A. Balanis, *Advanced Engineering Electromagnetics*, second edition, Wiley, 2012.

[3] General verification and validation practice for computational models: maintain separation between code verification, numerical diagnostics, reference comparison, uncertainty, and claim authorization.